FUTURE-READY LEARNING

Generative Artificial Intelligence (GEN AI)



Python • Prompt Engineering • LLMs • Vector Databases • RAG • Real-World AI Applications

A structured, hands-on program designed to build practical Generative AI skills - from fundamentals and programming to intelligent applications and deployment.

Industry-Focused
Career-oriented curriculum

Hands-On Learning
Projects and practical exercises

Career Support
Interview and placement guidance

LEARN • PRACTICE • BUILD • GROW

Build skills for an AI-powered future.

WHO WE ARE

About XpertsEdge Technologies

At XpertsEdge Technologies, we believe in shaping careers, not just teaching software. Our focus is on industry-ready skills, practical learning, expert guidance and career preparation.

Our training approach combines clear concept building with hands-on practice, real-time projects, interview preparation and mentoring. Learners are guided through the practical application of modern technologies so that classroom knowledge connects with real-world work.

What sets the learning experience apart is the emphasis on career growth - from understanding core concepts to building projects, strengthening portfolios and preparing for technical interviews.

01

Expert-led Learning

Concepts explained with practical and industry-oriented context.

02

Personalized Support

Guidance for learning, projects, interviews and career preparation.

03

Hands-On Practice

Real-world exercises and project-based learning.

04

Career-Focused Training

A learning path designed to make learners job-ready.

OUR APPROACH

Learn. Practice. Build. Grow.

A strong foundation, practical application and continuous career support form the core of the learning journey.

Real Skills
Real Projects
Career Readiness

THE OPPORTUNITY

Gateway to the Future: Generative AI Program

Generative AI with Python, Prompt Engineering, LLMs, Vector Databases & AI Applications is an advanced program designed for the next wave of digital transformation.

The supplied source brochure positions Generative AI as one of the fastest-growing technology sectors, reshaping industries, accelerating innovation and creating new career opportunities. Industries highlighted include IT, healthcare, finance, e-commerce, retail, manufacturing, cybersecurity and consulting.

$100B+
Market Size
Projected global Generative AI market value by 2030, as stated in the supplied source brochure.

2M+
Job Opportunities
Expected global demand for AI Engineers, Prompt Engineers & GenAI Specialists by 2027, as stated in the source brochure.

Global
Opportunities
The source highlights demand in the US, UK, Canada, Singapore and the EU for GenAI-related roles.

Career pathways highlighted in the source brochure

- AI — **AI Engineer**
- PE — **Prompt Engineer**
- GD — **GenAI Developer**
- AP — **AI Product Specialist**

Market and employment figures above are reproduced from the supplied source brochure for content completeness and should be independently verified before public publication.

WHY THIS LEARNING PATH

Why Choose XpertsEdge for GEN AI?

A practical GenAI learning path combining Python, Machine Learning concepts, LLMs, Prompt Engineering, projects and career preparation.

01

Industry-Relevant Skills

Learn modern GenAI concepts and tools that connect directly with real use cases.

02

Hands-On Projects

Move beyond theory through guided exercises, mini projects and a capstone project.

03

Expert Guidance

Learn with practical mentoring and explanations focused on application, not memorization.

04

Career Preparation

Mock interviews, portfolio development, resume guidance and interview preparation.

05

Real-World Focus

Build chatbots, RAG workflows, content tools and AI-powered applications.

06

Deployment Basics

Understand FastAPI/Flask, Streamlit/Gradio, Docker basics and introductory cloud usage.

Performance claims from the supplied source brochure - verify before publication

120%

Highest Hike

Unprecedented salary growth for learners transitioning into AI, ML & LLM engineering roles.

₹52 Lakhs

Highest Salary

Top placement package stated for GEN AI graduates in leading global tech and AI-driven companies.

65%

Industry Demand Growth

Explosive rise in GEN AI job roles across IT, finance, healthcare, automation, product development & research.

180+

Hiring Partners

A network of companies actively recruiting AI Engineers, Prompt Engineers & ML Developers.

120+

Top Industry Experts

Learn directly from industry professionals with years of real-world project experience.

100%

Job Placement

Career support stated as mock interviews, portfolio building, capstone projects and 1:1 mentoring.

STEP-BY-STEP PATH

Academic Journey

A progression from fundamentals and programming to real-world AI applications, projects and interview preparation.

01 AI & GenAI Fundamentals
Foundation concepts and real-world GenAI use cases.

02 Python for GenAI
Programming skills for AI applications.

03 Basic NLP
Text processing and language foundations.

04 Generative AI Core
LLMs, prompts and generative model concepts.

05 Embeddings & Vector Databases
Embeddings, semantic search and vector stores.

06 Retrieval-Augmented Generation
Connect LLMs with external knowledge.

07 GenAI Applications
Build practical GenAI applications.

08 Deployment & Basic MLOps
Deployment, APIs and MLOps basics.

09 Real-Time Projects
Apply learning through industry-style projects.

10 Mock Interviews
Practice technical and HR interview scenarios.

11 Live Interview Preparation / Placement
Prepare for interviews and career opportunities.

FROM LEARNING TO CAREER READINESS

Build a foundation. Apply it. Show it. Prepare to grow.

FUNDAMENTALS · PROJECTS · CAREER PREP

WHAT YOU WILL MASTER

Top Gen AI Skills You'll Master

01

AI & GenAI Fundamentals

Build a strong foundation in AI and Generative AI, understanding neural networks, transformers and modern LLMs like GPT-4. Learn core concepts, prompt engineering and responsible AI practices for real-world applications.

02

Python for GenAI

Develop Python skills for AI using libraries like NumPy, Pandas and PyTorch. Build, train and integrate GenAI models into practical applications.

03

Basic NLP

Understand text preprocessing, tokenization, embeddings and transformer-based models. Use tools like Hugging Face Transformers to build text classification and language understanding systems.

04

Generative AI Core

Learn how generative models like ChatGPT and Stable Diffusion create text and images. Master prompt design, fine-tuning and optimization for real-world use cases.

SKILL STACK

AI Fundamentals Python NLP LLMs Prompt Engineering

Learn • Practice • Build • Apply • Grow - practical GenAI learning with real-world project focus.

FROM MODELS TO APPLICATIONS

Build Intelligent Applications and Deploy Them

05

Embeddings & Vector Databases

Work with embeddings to power semantic search and intelligent retrieval systems. Use vector databases like Pinecone and FAISS for fast similarity search.

06

RAG (Retrieval-Augmented Generation)

Combine LLMs with external knowledge to improve response accuracy and reduce hallucinations. Build RAG pipelines using frameworks like LangChain for context-aware AI systems.

07

GenAI Applications

Create AI chatbots, content generators, recommendation systems and intelligent agents. Integrate AI capabilities using platforms like OpenAI API for scalable deployment.

08

Deployment & Basic MLOps

Learn deployment, containerization, CI/CD and model monitoring basics. Use tools like Docker and FastAPI to move from prototype to production.

Practical direction: Retrieval • Chatbots • APIs • Deployment • MLOps basics

Learn • Practice • Build • Apply • Grow - practical GenAI learning with real-world project focus.

CURRICULUM - PART 1

Fundamentals, Python and Basic NLP

AI & GenAI Fundamentals

- What is AI and Generative AI
- Types of AI (brief)
- Where GenAI is used in real projects

Required libraries

- NumPy
- Pandas
- Requests

Python for GenAI

- Python basics
- Variables, datatypes, conditions, loops
- Functions
- Strings, Lists, Dictionaries
- Basic file handling

Basic NLP

- What is NLP
- Text preprocessing
- Tokenization
- Cleaning text data

Foundation first

The curriculum starts with clear AI concepts and Python fundamentals before moving into language processing, models and applied GenAI.

CURRICULUM - PART 2

Generative AI Core

Large Language Models

- What are Large Language Models
- GPT overview
- LLaMA basics
- Falcon basics

Prompt Engineering

- Zero-shot
- Few-shot
- Chain-of-Thought

Model Adaptation

- Fine-tuning overview (concept only)
- LoRA and PEFT (high-level understanding)

CORE CONCEPTS

LLMs

Understand the models behind modern GenAI systems.

Prompts

Learn how instructions and examples shape model outputs.

Adaptation

Explore fine-tuning concepts and parameter-efficient techniques.

CURRICULUM - PART 3

Embeddings, Vector Databases and Hands-On GenAI

DB

Embeddings & Vector Databases

- What are embeddings
- Why vector databases are needed
- FAISS basics
- Chroma basics

Concept to retrieval

Learn how text is represented as vectors and how similarity search enables semantic retrieval.

AI

Hands-On GenAI

- Using OpenAI API
- Using Hugging Face models
- Building a chatbot step by step

Build while you learn

Connect APIs and models to create practical GenAI functionality, not just conceptual demos.

Learn • Practice • Build • Apply • Grow - practical GenAI learning with real-world project focus.

CURRICULUM - PART 4

RAG and Capstone Project

RAG

RAG (Retrieval-Augmented Generation)

- RAG concept
- Architecture flow
- Document ingestion
- Embedding creation
- Query + response flow

Combine LLMs with external knowledge so answers can use relevant retrieved context.

CP

Capstone Project

Company Knowledge Chatbot

- Upload PDFs / documents
- Ask questions using LLM + embeddings
- End-to-end RAG implementation

A practical end-to-end project that brings together ingestion, embeddings, retrieval and LLM responses.

Learn • Practice • Build • Apply • Grow - practical GenAI learning with real-world project focus.

CURRICULUM - PART 5

GenAI Applications and Mini Projects

GenAI Applications

Text to Text

- Summarization
- Q&A bots

Text to Image (overview)

- Stable Diffusion basics

Text to Code

- GitHub Copilot
- Code Llama

Mini Projects

1. **Resume / Interview Q&A Bot**
2. **Document Summarizer**
3. **Simple AI Image Generator**
4. **AI Content Tools**

Mini projects reinforce applied skills and help learners build portfolio-ready examples.

Learn • Practice • Build • Apply • Grow - practical GenAI learning with real-world project focus.

PROJECT CLASS EXTENSION + DEPLOYMENT

Machine Learning, Deep Learning & Deployment Basics

Strengthen the foundations behind modern Generative AI, then understand the practical path from a working prototype to a usable AI application.

ML

Machine Learning Basics

Covered in project class to connect learning from data, model behavior and traditional ML workflows with modern Generative AI systems.

DL

Deep Learning Basics

Neural-network foundations and the role deep learning plays in transformers and modern generative models.

Covered in Project Class

These concepts are connected to practical GenAI project work rather than treated as isolated theory.

API

API Serving

- FastAPI or Flask
- Expose application logic through APIs

UI

Simple AI Interfaces

- Streamlit or Gradio for UI
- Create usable front ends for GenAI demos

CL

Basic Cloud Intro

- AWS S3
- AWS Lambda
- Only usage - no deep cloud theory

OP

MLOps Basics

- Containerization concepts
- CI/CD introduction
- Monitoring basics

Practical objective: Understand how ML/DL foundations relate to GenAI and how a prototype can be served, given a simple UI, introduced to cloud services, and prepared for basic deployment workflows.

SUCCESS STORIES - SOURCE CONTENT REVIEW

Career Support and Outcome-Focused Preparation

The supplied source brochure included two named testimonials and a placement table with previous CTC, placed company and new CTC. Those records belonged to Aimore Technologies and have not been relabeled as XpertsEdge achievements.

Career Support Covered

- Mock interviews
- Resume guidance
- Portfolio building
- Capstone projects
- 1:1 mentoring / guidance
- Interview preparation

Original Source Page Included

- Testimonials from Shiva Sankar and Padma Priya
- Roles such as Senior Engineer AI and Senior Consultant
- Placed-at companies including Capgemini, Deloitte, Altimetrik and Digital Group
- Package figures ranging from 27 LPA to 52 LPA

PUBLISH-READY PRINCIPLE

Use XpertsEdge's own verified student results here.

This page preserves the structure and topic from the original 17-page brochure while avoiding attribution of another institute's student outcomes to XpertsEdge.

For a public marketing brochure, institute-specific testimonials and salary/placement claims should be supported by XpertsEdge's own verifiable records.

START YOUR GENERATIVE AI JOURNEY TODAY

Ready to Transform Your Career?

PROGRAM SNAPSHOT

2 Months
Duration

No Restrictions
Eligibility

Anyone passionate about building a career in Generative AI & Artificial Intelligence can enroll.

Who Can Join?

FRESHERS & GRADUATES

Launch your career in Generative AI with hands-on training in AI fundamentals, Machine Learning basics, LLMs, Prompt Engineering and popular GenAI tools. Work on text generation, chatbots, content creation and AI-powered applications.

WORKING PROFESSIONALS

Upskill or transition into roles such as AI Engineer, Prompt Engineer, AI Developer or Automation Specialist. Learn to design, build, fine-tune and deploy GenAI solutions and integrate AI into applications.

NON-IT PROFESSIONALS

Enter the AI and GenAI domain with confidence. Start from AI fundamentals, prompt engineering, tools and practical use cases through beginner-friendly, hands-on training.

Contact our Career Counselor • 8870783300 • info@xpertsedgetech.com

LET'S BUILD YOUR NEXT SKILL

Start Your GenAI Journey with XpertsEdge

Talk to us.

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LOCATION
Anna Nagar, Chennai

01

Learn

Build fundamentals with structured guidance.

02

Practice

Apply concepts through projects and hands-on exercises.

03

Grow

Prepare for interviews, strengthen your portfolio and pursue career opportunities.

BUILDING SUCCESS BEYOND BOUNDARIES

People • Skills • AI • A Better Tomorrow

The supplied source brochure listed Aimore branch contacts for Medavakkam, Anna Nagar, Porur and OMR. Those third-party branch details are intentionally not reproduced as XpertsEdge locations.